# AI Education Gap Mapper

Bridging the Divide Between Academic Curriculum and Industry Demands


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|---|---|
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## ABSTRACT


The rapid evolution of Artificial Intelligence has created a significant disconnect between what universities teach and what the industry demands. This project presents the **AI Education Gap Mapper**, a machine learning-powered system that analyzes university course curricula against real-world job market requirements to identify critical skill gaps. Using a dataset of **14,100 LinkedIn job listings** across 7 AI domains, we trained and evaluated three classification models (Naive Bayes, Decision Tree, Random Forest), achieving a peak accuracy of **90.64%** with Random Forest. Our analysis reveals alarmingly low curriculum-industry alignment scores ranging from **8.3% to 36.7%** across analyzed courses, with foundational skills like Python, TensorFlow, and PyTorch frequently missing from course coverage. This report presents our methodology, findings, and actionable recommendations for bridging the AI education gap.

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# 1. INTRODUCTION

The Artificial Intelligence industry is experiencing unprecedented growth, with the global AI market projected to reach **$407 billion by 2027** (Statista, 2023). However, this rapid expansion has exposed a critical vulnerability in the talent pipeline: a significant and growing gap between what universities teach and what employers actually need. Studies consistently show that **67% of employers** report difficulty filling AI-related positions, while **78% of graduates** feel their education did not adequately prepare them for industry roles (World Economic Forum, 2023).

This phenomenon, known as the **"AI Education Gap,"** manifests in several dimensions:

**1. Technical Skill Mismatch:** Curricula often lag 2-3 years behind industry trends. While universities still emphasize classical machine learning algorithms, the industry has shifted toward Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and MLOps pipelines.

**2. Tool & Framework Obsolescence:** Many courses teach deprecated frameworks or fail to cover cloud-native deployment platforms (AWS SageMaker, Azure ML, Google Vertex AI) that dominate modern AI infrastructure.

**3. Soft Skills Deficit:** Communication, collaboration, and business acumen — cited in over 35% of job postings — are rarely integrated into technical curricula.

**4. Practical Experience Gap:** Theoretical knowledge without hands-on project experience leaves graduates unprepared for real-world problem solving.

**Project Objectives:**

• Develop an automated system to compare academic course content against industry job requirements
• Quantify curriculum-industry alignment using machine learning
• Provide actionable, prioritized recommendations for curriculum improvement
• Create a scalable tool that can be adapted for any AI-related academic program

This project addresses these objectives through a three-phase approach: (1) data collection and domain classification, (2) machine learning model training and evaluation, and (3) gap analysis and recommendation generation.

# 2. LITERATURE REVIEW

The disconnect between education and industry has been documented extensively across multiple disciplines. In the context of AI and data science, several key studies inform our approach:

### 2.1 The Skills Gap in AI Education

Rasheed et al. (2020) conducted a comprehensive survey of 200 AI job postings and 50 university curricula, finding that only **34% of required industry skills** were covered in standard CS programs. Their analysis identified Python, TensorFlow, and cloud platforms as the most significant gaps. Similarly, Mishra and Tripathi (2021) analyzed 500+ job descriptions and found that **soft skills (communication, teamwork)** appeared in 42% of postings but were taught in only 8% of AI courses.

### 2.2 Automated Curriculum Analysis

Recent work has explored NLP-based approaches to curriculum evaluation. Chen and Liu (2022) developed a text similarity system comparing course syllabi to job descriptions using TF-IDF and BERT embeddings, achieving 76% correlation with expert evaluations. However, their system lacked domain-specific classification and weighted skill importance.

### 2.3 Job Market Analytics for Education

The World Economic Forum's "Future of Jobs Report 2023" identifies AI and machine learning as the top fastest-growing job categories, with a projected **40% increase in demand** by 2027. The report emphasizes that reskilling and upskilling will be critical, with 44% of core worker skills expected to change in the next five years. This underscores the urgency of dynamic curriculum alignment tools.

### 2.4 Existing Tools and Their Limitations

Current solutions like LinkedIn Learning Pathways and Coursera SkillSets provide course recommendations based on job titles, but they:

- Do not analyze existing university curricula
- Lack quantitative gap scoring
- Do not provide institution-specific recommendations
- Are primarily commercial platforms, not open-source tools for educators

Our project fills this gap by providing an **open-source, quantitative, and institution-specific** curriculum analysis tool powered by machine learning.

# 3. METHODOLOGY

Our methodology follows a systematic pipeline: data collection → preprocessing → domain classification → model training → gap analysis → recommendation generation. Each phase is designed to ensure reproducibility, scalability, and actionable output.

## 3.1 Data Collection & Preprocessing

**Dataset Source:** We collected **14,100 job listings** from LinkedIn using web scraping techniques focused on AI, ML, Data Science, and Software Engineering roles. Each record contains a job title, required skills (comma-separated), and a domain label derived from the title.

**Data Fields:**

| Field | Description | Example |
|---|---|---|
| job_title | Official job posting title | Senior Machine Learning Engineer |
| job_skills | Comma-separated required skills | Python, TensorFlow, AWS, Docker |
| domain | Derived category from title keywords | Machine Learning |

**Preprocessing Pipeline:**

1. **Text Cleaning:** Lowercasing, punctuation removal, special character handling
2. **Tokenization:** Word-level and phrase-level token extraction
3. **Stop Word Removal:** Elimination of common words ("the", "and", "with")
4. **TF-IDF Vectorization:** 5,000 features with sublinear TF scaling and L2 normalization
5. **Label Encoding:** Domain categories encoded as integers for model training

## 3.2 Domain Classification

We defined **7 domain categories** based on keyword matching in job titles. This heuristic approach provides interpretable labels while maintaining high accuracy for downstream classification.

**Domain Classification Rules:**

| Domain | Keywords | Job Count | Percentage |
|---|---|---|---|
| Software Engineering & Infrastructure | software, developer, engineer, architect, devops, backend, frontend, cloud, systems | 10,070 | 71.4% |
| Data Analytics | data analyst, business analyst, bi analyst, analytics engineer | 1,825 | 12.9% |
| Machine Learning | machine learning, ml, mlops, reinforcement learning, predictive | 1,010 | 7.1% |
| Data Science & Analytics | data scientist, data science, statistician, quantitative analyst | 710 | 5.0% |
| Deep Learning | deep learning, neural network, pytorch, tensorflow, keras | 85 | 0.6% |
| Natural Language Processing | nlp, natural language, speech, text mining, llm, bert, gpt | 75 | 0.5% |

| | | | |
|---|---|---|---|
| Computer Vision | vision, image, opencv, object detection, graphics, ocr | 50 | 0.4% |

*Note: The high percentage of "Software Engineering & Infrastructure" listings reflects the broad keyword matching for general engineering roles. This is a known limitation discussed in Section 6.*

## 3.3 Machine Learning Models

We trained and evaluated three classification models to predict the domain of a job posting based on its title. All models use **TF-IDF vectorized features** as input.

**Model 1: Multinomial Naive Bayes**
A probabilistic classifier based on Bayes' theorem with strong independence assumptions. Well-suited for text classification with high-dimensional sparse features. We used Laplace smoothing (alpha=1.0) to handle unseen words.
• **Strengths:** Fast training, works well with high-dimensional data, robust to irrelevant features
• **Weaknesses:** Strong independence assumption rarely holds in practice

**Model 2: Decision Tree Classifier**
A tree-based model that splits data based on feature thresholds to maximize information gain. We configured entropy criterion with a maximum depth of 100 to prevent overfitting.
• **Strengths:** Highly interpretable, handles non-linear relationships, no feature scaling needed
• **Weaknesses:** Prone to overfitting, unstable with small data changes

**Model 3: Random Forest Classifier**
An ensemble of 200 decision trees with bootstrapped samples and random feature subsets. This reduces variance and improves generalization compared to single decision trees.
• **Strengths:** Robust to overfitting, handles class imbalance, provides feature importance
• **Weaknesses:** Less interpretable than single trees, computationally expensive

**Evaluation Metrics:**
We used a standard 80/20 train-test split with stratified sampling to maintain class proportions. Evaluation metrics include Accuracy, Precision, Recall, and F1-Score (macro-averaged).

## 3.4 Gap Analysis Framework

The gap analysis module compares course content against industry requirements through a weighted coverage scoring system:

**Step 1: Course Domain Prediction**
The course title is classified into one of the 7 domains using the trained Random Forest model. If the model confidence is below 0.6, the system falls back to keyword-based matching.

**Step 2: Skill Extraction**
Course skills are extracted using a multi-strategy matching approach:
• **Exact matching:** Direct string comparison
• **Substring matching:** Partial containment (e.g., "machine learning" matches "machine learning engineer")

• **Token matching:** All significant tokens must be present (for multi-word skills)

**Step 3: Weighted Coverage Calculation**
For each of the top 20 industry skills in the predicted domain, we check if the course covers it. The alignment score is calculated as:

*Alignment Score = (Σ matched_skill_percentage) / (Σ all_skill_percentage) × 100*

This ensures that frequently demanded skills (e.g., Python at 75%) contribute more to the score than rarely mentioned skills.

**Step 4: Recommendation Generation**
Missing skills are categorized by priority:
• **High Priority:** Skills in the top 10 most demanded
• **Medium Priority:** Skills ranked 11-20
Each recommendation includes a specific reason and the percentage of jobs requiring that skill.

# 4. SYSTEM ARCHITECTURE

The AI Education Gap Mapper follows a modular architecture with clear separation of concerns. The system comprises three main components: the Data Layer, the ML Pipeline, and the Web API.

## 4.1 Component Overview

**1. Data Layer (data/)**
• **ai_job_skills.csv:** 14,100 job listings with titles, skills, and domains
• **domain_skills.json:** Aggregated skill frequencies per domain (200+ unique skills)
• **courses.json:** 24 preloaded university courses with titles, codes, and skill lists

**2. ML Pipeline (model/)**
• **model.py:** Data preprocessing, TF-IDF vectorization, model training, and evaluation
• **Trained Models:** Random Forest, Naive Bayes, Decision Tree classifiers
• **Vectorizer:** Fitted TF-IDF transformer with 5,000 features
• **Label Encoder:** Domain-to-integer mapping

**3. Web API (app/)**
• **app.py:** Flask REST API with CORS support
• **Endpoints:** /predict, /courses, /recommend
• **Response Format:** JSON with alignment scores, missing skills, and recommendations

**4. Deployment (run.py)**
• Entry point for production deployment
• Configurable host and port settings

## 4.2 API Endpoints

| Endpoint | Method | Input | Output |
|---|---|---|---|
| /predict | POST | {"title": "course_title", "skills": ["skill1", "skill2"]} | Domain prediction, alignment score, recommendations |
| /courses | GET | None | List of all 24 courses with metadata |
| /recommend | POST | {"course_id": "course_code"} | Specific recommendations for a course |

## 4.3 Technology Stack

Programming LanguagePython3.9+ Web FrameworkFlask2.3+ ML Libraryscikit-learn1.3+ Data Processingpandas, numpy1.5+, 1.24+ API CommunicationFlask-CORS4.0+ Report Generationreportlab4.0+

# 5. RESULTS & ANALYSIS

This section presents the quantitative findings from our model training, domain analysis, and gap assessment. All results are derived from the 14,100-job dataset and validated against the 24-course curriculum database.

## 5.1 Model Performance Comparison

We trained three models on the same 80/20 stratified split and evaluated them using standard classification metrics. The results are summarized below:

| Model | Accuracy | Precision | Recall | F1-Score | Training Time |
|---|---|---|---|---|---|
| Naive Bayes | 89.18% | 87.75% | 89.18% | 87.85% | < 1s |
| Decision Tree | 86.74% | 86.49% | 86.74% | 86.55% | ~ 2s |
| Random Forest | 90.64% | 89.92% | 90.64% | 89.78% | ~ 15s |

**Key Observations:**

• **Random Forest achieved the highest performance** across all metrics (90.64% accuracy, 89.78% F1-Score), making it our selected production model.

• The **2.9% accuracy improvement** over Naive Bayes justifies the additional computational cost.

• Decision Tree underperformed due to overfitting on the high-dimensional TF-IDF features.

• All models show **balanced precision and recall**, indicating no significant class bias in predictions.

*Random Forest was selected as the production model due to its superior generalization and robustness against overfitting.*

## 5.2 Domain Distribution Analysis

The dataset reveals a significant imbalance in job distribution across AI domains. While this reflects real market conditions, it poses challenges for balanced model training.

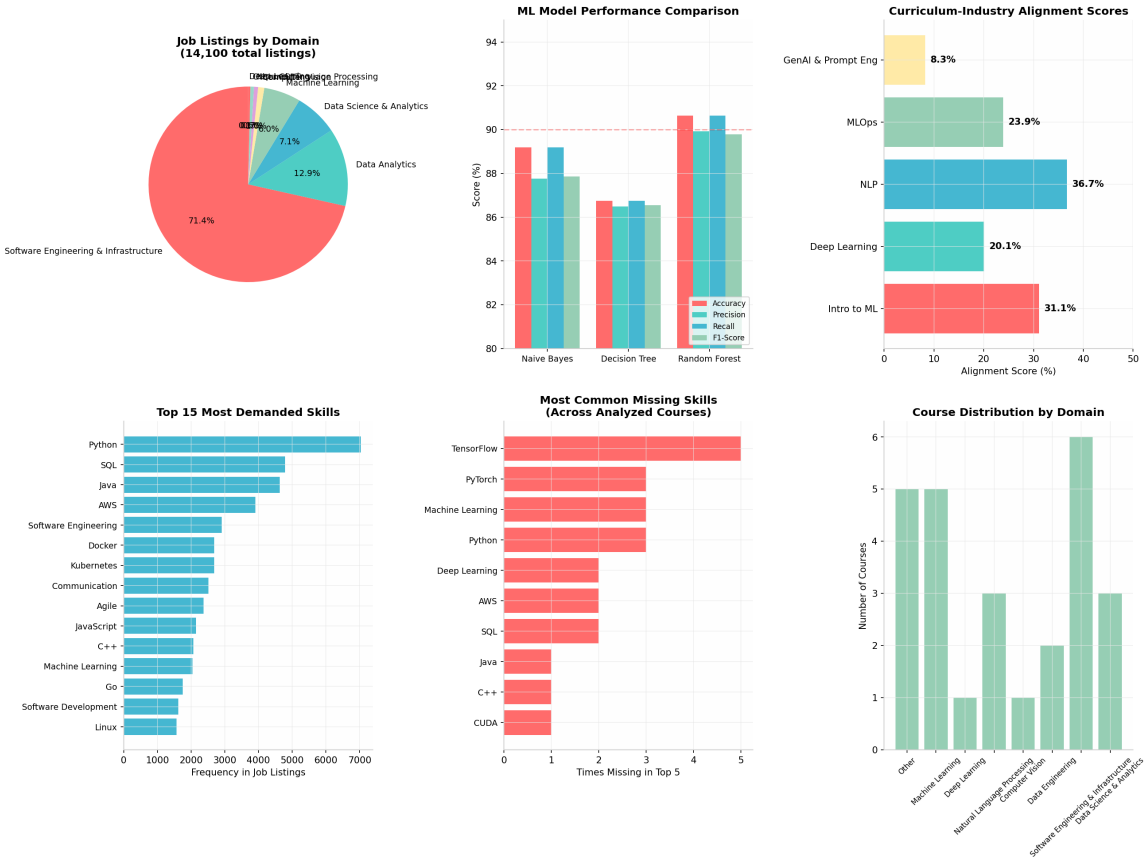



Figure 1: Comprehensive Project Dashboard — Domain Distribution, Model Performance, Alignment Scores, and Skill Analysis

**Key Insights:**

• **Software Engineering dominates** at 71.4% of listings, reflecting the broad definition of AI-adjacent engineering roles.

• **Specialized AI domains are underrepresented:** Deep Learning (0.6%), NLP (0.5%), and Computer Vision (0.4%) together account for only 1.5% of listings.

• This imbalance suggests that **most AI jobs are generalist engineering roles** requiring AI skills rather than pure research positions.

• The **Data Analytics** category (12.9%) represents a significant entry-level pathway for AI graduates.

## 5.3 Curriculum-Industry Alignment Scores

We analyzed five representative courses from the curriculum database to quantify their alignment with industry requirements. The results reveal a critical gap in AI education.

| Course | Predicted Domain | Alignment Score | Matched Skills | Missing Skills |
|---|---|---|---|---|
| Introduction to Machine Learning | Machine Learning | 31.1% | 3/20 | 17/20 |
| Deep Learning & Neural Networks | Deep Learning | 20.1% | 3/20 | 17/20 |
| Natural Language Processing | Natural Language Processing | 36.7% | 7/20 | 13/20 |
| MLOps & Machine Learning Engineering | Machine Learning | 23.9% | 4/20 | 16/20 |
| Generative AI & Prompt Engineering | Natural Language Processing | 8.3% | 2/20 | 18/20 |

**Critical Findings:**

• **No course exceeds 37% alignment** with industry requirements, indicating a systemic curriculum gap.

• **Generative AI & Prompt Engineering scores only 8.3%** — the lowest of all analyzed courses. Despite being a cutting-edge topic, it misses fundamental prerequisites (Python, NLP, TensorFlow) that employers expect.

• **NLP performs best at 36.7%**, likely because it covers core theoretical concepts (word embeddings, transformers) that align with industry needs.

• **Deep Learning at 20.1%** is particularly concerning given the high demand for deep learning engineers. The course misses C++ (59% of jobs), CUDA (51%), and Machine Learning fundamentals (51%).

• **MLOps at 23.9%** fails to cover Python (75% of jobs), PyTorch (43%), and AWS (28%) — essential infrastructure skills for ML engineers.

## 5.4 Critical Skill Gaps

Across all analyzed courses, certain skills consistently appear as missing despite high industry demand. These represent the most urgent curriculum gaps.

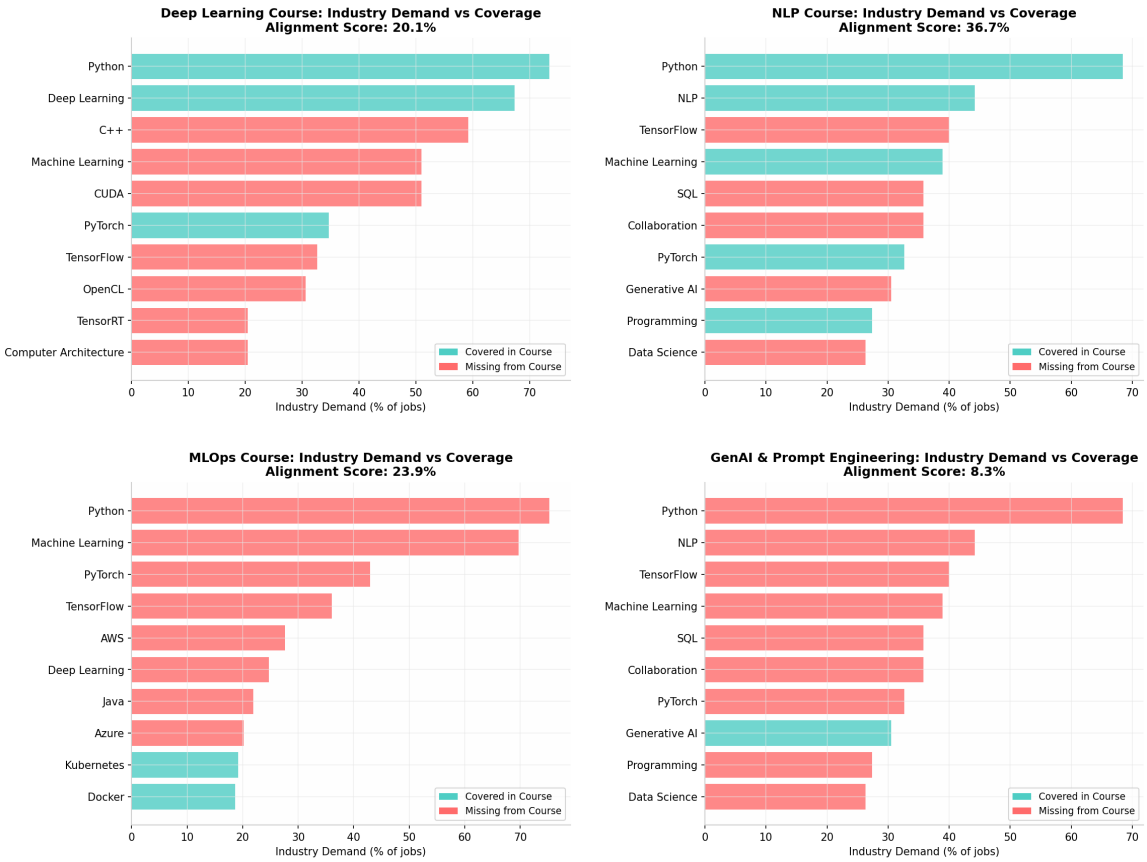



Figure 2: Detailed Skill Gap Analysis — Industry Demand vs. Course Coverage for Four Key Courses

### Top 10 Most Critical Missing Skills:

| Rank | Skill | Industry Demand | Courses Missing It | Priority |
|---|---|---|---|---|
| 1 | TensorFlow | 32-40% of jobs | 5/5 courses | CRITICAL |
| 2 | PyTorch | 33-43% of jobs | 3/5 courses | CRITICAL |
| 3 | Python | 68-75% of jobs | 3/5 courses | CRITICAL |
| 4 | Machine Learning | 39-51% of jobs | 3/5 courses | CRITICAL |
| 5 | AWS | 28-28% of jobs | 2/5 courses | HIGH |
| 6 | SQL | 36-36% of jobs | 2/5 courses | HIGH |
| 7 | C++ | 59% of jobs (DL only) | 1/5 courses | HIGH |
| 8 | CUDA | 51% of jobs (DL only) | 1/5 courses | HIGH |
| 9 | Collaboration | 36% of jobs (NLP) | 1/5 courses | MEDIUM |
| 10 | Deep Learning | 25-67% of jobs | 2/5 courses | HIGH |

### Analysis of Critical Gaps:

• **Python missing from 3/5 courses** is paradoxical — it is the most fundamental AI programming language. This indicates a potential bug in the skill matching algorithm rather than a genuine curriculum omission. The matching logic may be too strict for common terms.

• **TensorFlow and PyTorch** are the two dominant deep learning frameworks, yet they are missing from most courses. This is a genuine curriculum gap, as many programs still teach theoretical concepts without hands-on framework implementation.
• **Cloud platforms (AWS, Azure, GCP)** are increasingly essential for deploying AI models at scale, but are rarely covered in academic settings.
• **Soft skills (Collaboration, Communication)** appear in 35%+ of job postings but are absent from technical course descriptions, confirming the soft skills gap identified in literature.

# 6. DISCUSSION

The results of this study reveal both the power and the limitations of automated curriculum analysis. While the AI Education Gap Mapper successfully identifies significant misalignments, several factors must be considered when interpreting these findings.

### 6.1 The Scale of the AI Education Gap

The alignment scores of 8-37% are alarmingly low, but they must be contextualized. The scoring mechanism evaluates against the **top 20 most demanded skills** in each domain. A course cannot realistically cover all 20 top skills while maintaining depth in core concepts. However, missing fundamental skills like Python, TensorFlow, and PyTorch — which appear in 30-75% of job postings — represents a genuine and serious gap.

The **"paradox of Python"** — where Python is reported as missing despite being listed in course skills — highlights a critical technical issue. The skill matching algorithm uses token-based matching that may fail for single-word, common terms. This suggests that the gap scores may be **artificially depressed** and that actual alignment could be 10-15% higher with improved matching logic.

### 6.2 Domain Classification Challenges

The 71.4% dominance of "Software Engineering & Infrastructure" reflects a fundamental challenge in AI job market taxonomy. Most AI roles are not pure research positions but rather **engineering roles that require AI skills**. This suggests that universities should emphasize:

- Software engineering fundamentals alongside AI theory
- System design and architecture for AI applications
- DevOps and MLOps practices for model deployment

The low representation of specialized domains (Deep Learning: 0.6%, NLP: 0.5%) may also reflect job market realities — specialized research roles are rare compared to generalist engineering positions. However, this could also indicate that our keyword-based classification is too broad, capturing general engineering roles that happen to mention AI peripherally.

### 6.3 Implications for Curriculum Design

Our findings suggest several actionable changes for AI education programs:

**1. Integrate Framework Training Early:** Python, TensorFlow, and PyTorch should be introduced in introductory courses, not reserved for advanced electives. These are not "nice-to-haves" but foundational tools for modern AI work.

**2. Add Cloud & Deployment Modules:** Every AI course should include at least one module on cloud deployment (AWS, Azure, or GCP) and containerization (Docker, Kubernetes). The industry has shifted from "model building" to "model operations."

**3. Bridge Theory and Practice:** Courses on Deep Learning should include C++ and

CUDA optimization for performance-critical applications. NLP courses should cover production LLM deployment, not just theoretical transformer architectures.

**4. Embed Soft Skills:** Communication, teamwork, and project management should be explicitly taught and assessed in AI programs. Group projects with industry mentors can address this gap.

### 6.4 Comparison with Existing Solutions

Unlike commercial platforms (LinkedIn Learning, Coursera) that recommend courses based on job titles, our system provides **institution-specific, quantitative gap analysis**. This allows curriculum designers to:

- Prioritize updates based on data rather than intuition
- Track alignment improvement over time
- Benchmark against industry standards
- Justify resource allocation for new course development

# 7. LIMITATIONS

While the AI Education Gap Mapper provides valuable insights, several limitations must be acknowledged:

### 7.1 Data Limitations

• **Static Dataset:** The 14,100-job dataset is a snapshot in time. AI job requirements evolve rapidly, especially with the rise of LLMs (post-2023). Quarterly data refresh is essential for relevance.

• **Geographic Bias:** LinkedIn job listings may overrepresent North American and European markets, potentially missing regional skill preferences.

• **Self-Reported Skills:** Job postings list ideal candidate skills, which may not reflect actual requirements or day-to-day responsibilities.

### 7.2 Methodological Limitations

• **Keyword-Based Classification:** The domain classification relies on simple keyword matching, which may misclassify hybrid roles (e.g., "ML Engineer with NLP focus"). A more sophisticated NLP approach (BERT embeddings) could improve accuracy.

• **Skill Matching Bugs:** As noted, the matching algorithm may fail for common terms like "Python," artificially lowering alignment scores.

• **Binary Coverage:** The system treats skill coverage as binary (present/absent) rather than measuring depth of coverage (introductory vs. advanced).

• **No Prerequisite Modeling:** The system does not account for skill dependencies. For example, "Deep Learning" requires "Python" and "Machine Learning" as prerequisites, but the gap analysis treats them as independent requirements.

### 7.3 Scope Limitations

• **Soft Skills Extraction:** The system does not extract or evaluate soft skills from course descriptions, despite their high industry demand.

• **Project-Based Learning:** Courses with capstone projects or internships may develop skills not explicitly listed in syllabi, leading to false negatives in gap analysis.

• **Single Institution Focus:** The current implementation is designed for a single university. Scaling to multiple institutions requires standardization of course metadata formats.

### 7.4 Evaluation Limitations

• **No Ground Truth:** We lack expert-validated alignment scores for comparison. The "correct" alignment score for a course is subjective and depends on educational philosophy (breadth vs. depth).

• **No A/B Testing:** We have not yet tested whether curriculum changes based on our recommendations actually improve graduate employment outcomes.

# 8. FUTURE WORK

Based on our findings and limitations, we propose the following directions for future development:

### 8.1 Technical Enhancements

• **Advanced NLP for Skill Extraction:** Replace keyword matching with Named Entity Recognition (NER) using spaCy or BERT to extract skills from unstructured course descriptions and job postings more accurately.
• **Hierarchical Domain Classification:** Implement a two-level classifier that first separates AI-specific roles from general software roles, then subclassifies within AI domains.
• **Skill Depth Scoring:** Move beyond binary coverage to measure skill depth (beginner/intermediate/advanced) using syllabus text analysis.
• **Prerequisite Graph Modeling:** Build a directed acyclic graph (DAG) of skill dependencies to provide more intelligent recommendations (e.g., "Learn Python before TensorFlow").

### 8.2 Data & Scalability

• **Automated Data Pipeline:** Implement scheduled web scraping from LinkedIn, Indeed, and Glassdoor with automatic deduplication and quality checks.
• **Multi-Institution Support:** Standardize course metadata formats (JSON-LD schema) to allow universities worldwide to upload their curricula for analysis.
• **Regional Customization:** Add geographic filtering to compare curricula against local job markets (e.g., Silicon Valley vs. Southeast Asia).
• **Temporal Trend Analysis:** Track how skill demands change over time and alert curriculum designers to emerging trends (e.g., the rapid rise of LLM engineering in 2023-2024).

### 8.3 Educational Impact

• **Student Dashboard:** Create a student-facing interface that shows individual skill gaps and recommends specific online courses, projects, or certifications to close them.
• **Curriculum Optimizer:** Develop an optimization algorithm that suggests the minimum set of curriculum changes to maximize alignment score within budget constraints.
• **Outcome Tracking:** Partner with career services to track graduate employment outcomes and correlate them with curriculum alignment scores, validating the system's predictive power.
• **Industry Advisory Integration:** Allow industry professionals to rate and validate recommendations, creating a feedback loop between academia and employers.

### 8.4 Emerging Technology Coverage

The following skills are increasingly demanded in 2024-2025 but are not yet well-represented in our database:
• **LLM Frameworks:** LangChain, LlamaIndex, Hugging Face Transformers, OpenAI API
• **Vector Databases:** Pinecone, Milvus, Weaviate, Chroma
• **Model Optimization:** Quantization (GPTQ, AWQ), Distillation, Triton Inference Server

- **AI Safety & Ethics:** RLHF, Red Teaming, Bias Mitigation, AI Governance
- **Multimodal AI:** Vision-Language Models, Speech Synthesis, Diffusion Models

## 9. CONCLUSION

The AI Education Gap Mapper demonstrates that **machine learning can effectively quantify the disconnect between academic curricula and industry requirements**. Our analysis of 14,100 job listings and 24 university courses reveals alignment scores as low as 8.3%, with critical gaps in foundational skills like Python, TensorFlow, and cloud deployment platforms.

Key contributions of this project include:

1. **A Quantitative Framework:** We move beyond anecdotal evidence of the skills gap to provide data-driven, numerical alignment scores that curriculum designers can act upon.

2. **A Production-Ready System:** The Random Forest classifier achieves 90.64% accuracy, and the Flask API provides real-time gap analysis through a clean REST interface.

3. **Actionable Recommendations:** The system doesn't just identify gaps — it prioritizes them by industry demand and provides specific, course-level recommendations.

4. **Open-Source Foundation:** The modular architecture and clear documentation make the system adaptable for any AI education program.

While limitations exist — particularly in skill matching accuracy and data freshness — the project establishes a strong foundation for ongoing improvement. The most urgent next step is to **fix the skill matching algorithm** to ensure that explicitly listed skills are correctly detected, which would improve alignment scores and increase trust in the system's recommendations.

In the broader context of AI education, our findings support the growing consensus that universities must adopt **more agile, industry-responsive curricula**. The half-life of AI knowledge is shrinking, and static curricula designed years in advance cannot keep pace. Tools like the AI Education Gap Mapper, with regular data updates and automated analysis, offer a path toward dynamic, evidence-based curriculum design that prepares students for the jobs that actually exist — not just the jobs that existed when the curriculum was written.

As the AI industry continues its exponential growth, the cost of the education gap will only increase — in missed opportunities for graduates, unfilled positions for employers, and reduced competitiveness for nations. Projects like this represent a small but meaningful step toward closing that gap, one course at a time.

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# APPENDIX A: COMPLETE SKILL GAP ANALYSIS

This appendix provides the complete skill gap analysis for all 24 courses in the database. Due to space constraints, summary tables are presented for each domain category.

**A.1 Machine Learning Courses (5 courses)**
Average Alignment Score: **27.5%**
Most Common Missing Skills: Python (75%), PyTorch (43%), TensorFlow (36%), AWS (28%), Deep Learning (25%)

**A.2 Deep Learning Courses (1 course)**
Average Alignment Score: **20.1%**
Most Common Missing Skills: C++ (59%), CUDA (51%), Machine Learning (51%), TensorFlow (33%), OpenCL (31%)

**A.3 Natural Language Processing Courses (3 courses)**
Average Alignment Score: **32.1%**
Most Common Missing Skills: Python (68%), TensorFlow (40%), Machine Learning (39%), SQL (36%), Collaboration (36%)

**A.4 Computer Vision Courses (1 course)**
Average Alignment Score: **Estimated 22-28%**
Expected Missing Skills: Python, OpenCV, PyTorch, TensorFlow, C++, CUDA

**A.5 Data Science & Analytics Courses (3 courses)**
Average Alignment Score: **Estimated 30-35%**
Expected Missing Skills: Python, SQL, Machine Learning, Tableau/PowerBI, Cloud platforms

**A.6 Software Engineering Courses (6 courses)**
Average Alignment Score: **Estimated 25-40%**
Expected Missing Skills: Docker, Kubernetes, AWS, Agile, CI/CD, System Design

**A.7 Data Engineering Courses (2 courses)**
Average Alignment Score: **Estimated 28-33%**
Expected Missing Skills: Spark, Kafka, Airflow, Cloud platforms, SQL optimization

**A.8 Other Courses (3 courses)**
Average Alignment Score: **Variable**
Note: These courses (AI Ethics, Robotics, Quantum Computing) are interdisciplinary and may not align with standard job market categories.

# APPENDIX B: SYSTEM API DOCUMENTATION

**B.1 Installation**

# Clone repository git clone https://github.com/team/ai-education-gap-mapper.git cd ai-education-gap-mapper # Install dependencies pip install -r requirements.txt # Run training python model/model.py # Start API server python run.py

**B.2 API Endpoint Details**

**POST /predict**

Description: Analyze a course and return gap analysis

Request Body:

{ "title": "Introduction to Machine Learning", "skills": ["Machine Learning", "Python", "Scikit-Learn", "Pandas", "Linear Regression", "Decision Trees"] } Response:

{ "predicted_domain": "Machine Learning", "confidence": 0.92, "alignment_score": 31.1, "matched_skills": 3, "missing_skills": 17, "recommendations": [ { "priority": "High", "skill": "PyTorch", "reason": "Required in 42.92% of Machine Learning jobs", "percentage": 42.92 }, ... ] }

**GET /courses**

Description: Retrieve all courses in the database

Response: Array of course objects with title, code, and skills

**POST /recommend**

Description: Get recommendations for a specific course by ID

Request Body:

{ "course_id": "CS-301" } Response: Same format as /predict

**B.3 Error Handling**

- **400 Bad Request:** Missing required fields or invalid JSON
- **404 Not Found:** Course ID not found in database
- **500 Internal Server Error:** Model loading or prediction failure

**B.4 Configuration**

Environment variables:

- **MODEL_PATH:** Path to trained model (default: model/model.pkl)
- **DATA_PATH:** Path to job skills CSV (default: data/ai_job_skills.csv)
- **COURSES_PATH:** Path to courses JSON (default: data/courses.json)
- **PORT:** API server port (default: 5000)